INCH-POUND

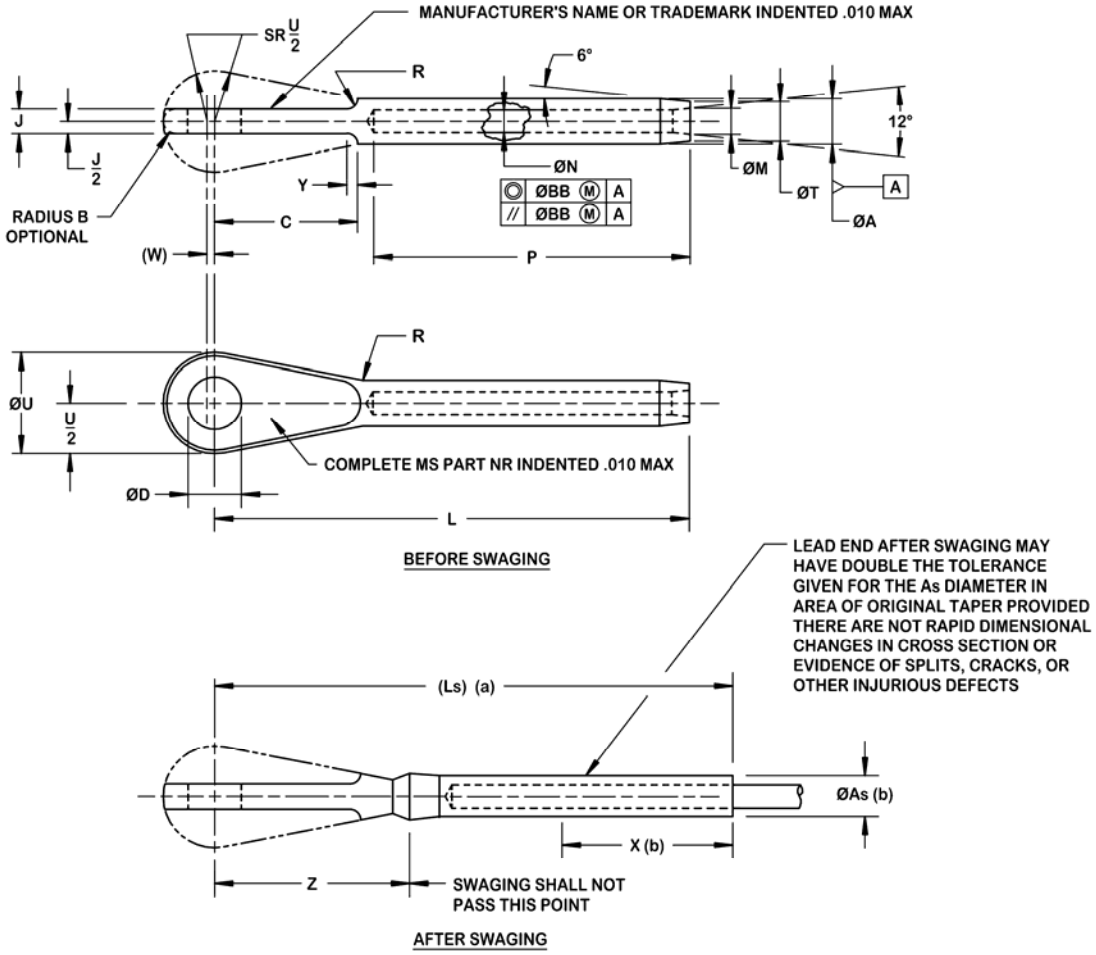
MS20668G
w/AMENDMENT 2
8 March 2011
SUPERSEDING
MS20668G
w/AMENDMENT 1
8 November 2006

DETAIL SPECIFICATION SHEET

# TERMINAL, WIRE ROPE, SWAGING, EYE END

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet, MIL-DTL-781, and QPL-781.




NOTES: (a) Reference dimensions are for design purposes only and are not an inspection requirement.
(b) Swaged terminals shall conform to ∅As for length X.

FIGURE 1. Terminal, wire rope, swaging, eye end.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:23Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and dimensions.

| Dash number | Wire rope diameter | | Minimum breaking strength lb 1/ | ØA | | ØAs | | B radius | C ±.020 |
|---|---|---|---|---|---|---|---|---|---|
| | Nominal reference | Minimum | | | | | | | |
| -2 | 1/16 | 0.062 | 480 | 0.160 | +0.000 -0.005 | 0.138 | +0.000 -0.005 | 0.062 | 0.523 |
| -3 | 3/32 | 0.093 | 920 | 0.218 | | 0.190 | | 0.078 | 0.707 |
| -4 | 1/8 | 0.125 | 2,000 | 0.250 | | 0.219 | | 0.125 | 0.738 |
| -5 | 5/32 | 0.156 | 2,800 | 0.297 | | 0.250 | | 0.140 | 0.831 |
| -6 | 3/16 | 0.187 | 4,200 | 0.359 | | 0.313 | | 0.171 | 0.903 |
| -7 | 7/32 | 0.218 | 5,600 | 0.427 | | 0.375 | +0.000 -0.007 | 0.187 | 1.007 |
| -8 | 1/4 | 0.250 | 7,000 | 0.494 | | 0.438 | | 0.203 | 1.133 |
| -9 | 9/32 | 0.281 | 8,000 | 0.563 | | 0.500 | +0.000 -0.008 | | 1.257 |
| -10 | 5/16 | 0.312 | 9,800 | 0.635 | | 0.563 | | 0.218 | 1.373 |
| -12 | 3/8 | 0.375 | 14,400 | 0.703 | +0.000 -0.005 | 0.625 | | 0.255 | 1.688 |
| -14 | 7/16 | 0.437 | 17,600 | 0.781 | | 0.688 | | 0.275 | 1.968 |
| -16 | 1/2 | 0.500 | 22,800 | 0.844 | | 0.750 | +0.000 -0.009 | 0.295 | 2.115 |
| -18 | 9/16 | 0.562 | 28,500 | 0.984 | | 0.875 | | 0.310 | 2.625 |
| -20 | 5/8 | 0.625 | 35,000 | 1.109 | | 1.000 | +0.000 -0.010 | 0.321 | 3.062 |
| -24 | 3/4 | 0.750 | 49,600 | 1.359 | | 1.250 | +0.000 -0.012 | 0.345 | 3.093 |
| -28 | 7/8 | 0.875 | 66,500 | 1.593 | +0.000 -0.010 | 1.437 | | 0.360 | 3.515 |
| -32 | 1 | 1.000 | 85,400 | 1.812 | | 1.625 | | 0.375 | 3.937 |

1/ To achieve the minimum breaking strength, for the terminal test only, a galvanized carbon steel wire rope shall be used.

TABLE I. Dash numbers and dimensions - Continued.

| Dash number | ØD | | J | | L +0.020 -0.000 | Ls reference | ØM | |
|---|---|---|---|---|---|---|---|---|
| -2 | 0.190 | +0.002 -0.000 | 0.088 | +0.000 -0.005 | 1.631 | 1.809 | 0.090 | +0.010 -0.000 |
| -3 | | | 0.103 | | 2.043 | 2.160 | 0.119 | |
| -4 | | | 0.190 | | 2.337 | 2.593 | 0.154 | |
| -5 | 0.250 | | 0.197 | | 2.684 | 3.029 | 0.188 | |
| -6 | 0.313 | | 0.255 | | 3.019 | 3.187 | 0.223 | |
| -7 | | | 0.291 | | 3.382 | 3.678 | 0.257 | |
| -8 | 0.375 | | 0.307 | | 3.763 | 4.062 | 0.291 | |
| -9 | 0.438 | | 0.322 | | 4.153 | 4.512 | 0.326 | |
| -10 | | | 0.343 | | 4.546 | 4.969 | 0.360 | |
| -12 | 0.500 | +0.005 -0.000 | 0.375 | +0.000 -0.015 | 5.562 | 5.968 | 0.430 | +0.012 -0.000 |
| -14 | 0.562 | | | | 6.398 | 6.867 | 0.514 | |
| -16 | 0.625 | | 0.468 | | 7.323 | 7.886 | 0.584 | |
| -18 | 0.750 | | 0.562 | | 8.185 | 8.778 | 0.653 | |
| -20 | 0.875 | | 0.657 | | 9.167 | 9.854 | 0.722 | |
| -24 | | | | | 10.328 | 10.900 | 0.860 | +0.015 -0.000 |
| -28 | 1.000 | | 0.750 | | 11.530 | 12.290 | 1.013 | |
| -32 | 1.125 | | 0.844 | | 13.156 | 14.062 | 1.151 | |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:23Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and dimensions - Continued.

| Dash number | ØN | | P | | R radius reference | ØT | |
|---|---|---|---|---|---|---|---|
| -2 | 0.078 | +0.005 -0.000 | 1.042 | +0.031 -0.000 | 0.088 | 0.136 | +0.000 -0.005 |
| -3 | 0.109 | | 1.261 | | 0.103 | 0.190 | |
| -4 | 0.141 | | 1.511 | | 0.190 | 0.219 | |
| -5 | 0.172 | | 1.761 | | 0.197 | 0.250 | |
| -6 | 0.203 | | 2.011 | | 0.255 | 0.313 | |
| -7 | 0.234 | | 2.261 | | 0.291 | 0.375 | +0.000 -0.007 |
| -8 | 0.265 | | 2.511 | | 0.307 | 0.438 | |
| -9 | 0.297 | | 2.761 | | 0.322 | 0.500 | +0.000 -0.008 |
| -10 | 0.328 | | 3.011 | | 0.343 | 0.563 | |
| -12 | 0.390 | +0.008 -0.000 | 3.511 | +0.047 -0.000 | 0.375 | 0.625 | |
| -14 | 0.468 | | 4.011 | | | 0.688 | |
| -16 | 0.531 | +0.009 -0.000 | 4.698 | | 0.468 | 0.750 | +0.000 -0.009 |
| -18 | 0.594 | | 5.011 | | 0.562 | 0.875 | |
| -20 | 0.656 | +0.010 -0.000 | 5.511 | +0.062 -0.000 | 0.657 | 1.000 | +0.000 -0.010 |
| -24 | 0.781 | +0.012 -0.000 | 6.511 | | | 1.250 | +0.000 -0.012 |
| -28 | 0.921 | | 7.166 | | 0.750 | 1.437 | |
| -32 | 1.046 | | 8.229 | | 0.844 | 1.625 | |

TABLE I. Dash numbers and dimensions - Continued.

| Dash number | ØU +0.025 -0.010 | W Reference | X Minimum | Y | Z | ØBB (FIM) |
|---|---|---|---|---|---|---|
| -2 | 0.359 | 0.031 | 0.70 | 0.072 | 0.662 | 0.016 |
| -3 | 0.438 | | 0.80 | 0.092 | 0.856 | |
| -4 | 0.500 | | 1.05 | 0.103 | 0.900 | |
| -5 | 0.640 | | 1.29 | 0.131 | 0.997 | |
| -6 | 0.781 | 0.047 | 1.31 | 0.155 | 1.082 | |
| -7 | 0.813 | | 1.55 | 0.187 | 1.195 | |
| -8 | 0.968 | | 1.70 | 0.221 | 1.326 | |
| -9 | 1.109 | 0.063 | 1.89 | 0.251 | 1.465 | |
| -10 | 1.218 | | 2.06 | 0.281 | 1.609 | |
| -12 | 1.500 | | 3.12 | 0.320 | 2.124 | 0.020 |
| -14 | 1.750 | | 3.57 | 0.350 | 2.460 | |
| -16 | 1.875 | | 4.31 | 0.406 | 2.698 | |
| -18 | 2.340 | | 4.51 | 0.450 | 3.247 | |
| -20 | 2.730 | | 5.04 | 0.515 | 3.729 | 0.030 |
| -24 | 2.750 | 0.093 | 5.80 | 0.600 | 3.890 | |
| -28 | 3.125 | | 6.31 | 0.675 | 4.437 | 0.040 |
| -32 | 3.510 | | 7.26 | 0.775 | 5.000 | |

REQUIREMENTS:

1. Material: Material shall be in accordance with MIL-DTL-781.

2. Finish: Finish shall be in accordance with MIL-DTL-781.

3. Swage: Swage shall be in accordance with MIL-DTL-6117.

4. Tolerances: Unless otherwise specified, tolerances: decimals ± 0.010, angles ± 3°.

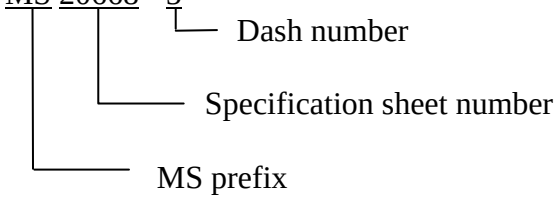


Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:23Z
Check the source to verify that this is the current version before use.

NOTES:

1. The part or identifying number (PIN) consists of the letters MS, the specification sheet number, and a dash number taken from table I.

EXAMPLE: MS 20668 - 5




2. Dimensions are in inches.

3. Remove burrs and sharp edges. (See MIL-DTL-781.)

4. Interpret drawing in accordance with ASME Y14.5.

5. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence.

6. Unless otherwise specified, issues of referenced documents are those in effect at the time of solicitation.

7. Corrosion resistant steel parts can universally replace carbon and alloy steel parts as shown in table II. Carbon and alloy steel parts are inactive for new design and cannot be substituted for corrosion resistant steel parts.

TABLE II. Substitution table.

| MS part numbers | |
|---|---|
| Corrosion resistant steel | Carbon steel, cadmium plated |
| MS20668-2 | MS20668F2 |
| MS20668-3 | MS20668F3 |
| MS20668-4 | MS20668F4 |
| MS20668-5 | MS20668F5 |
| MS20668-6 | MS20668F6 |
| MS20668-7 | MS20668F7 |
| MS20668-8 | MS20668F8 |
| MS20668-9 | MS20668F9 |
| MS20668-10 | MS20668F10 |
| MS20668-12 | MS20668F12 |
| MS20668-14 | MS20668F14 |
| MS20668-16 | MS20668F16 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:23Z
Check the source to verify that this is the current version before use.

TABLE II. Substitution table - Continued.

| MS part numbers | |
|---|---|
| Corrosion resistant steel | Carbon steel, cadmium plated |
| MS20668-18 | MS20668F18 |
| MS20668-20 | MS20668F20 |
| MS20668-24 | MS20668F24 |
| MS20668-28 | MS20668F28 |
| MS20668-32 | MS20668F32 |

AMENDMENT NOTATIONS: The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR4
Navy - AS
Air Force - 99
DLA - GS

Review Activity:
Air Force - 71

Preparing Activity:
DLA - GS5

(Project 1640-2011-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.daps.dla.mil/.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:23Z
Check the source to verify that this is the current version before use.